Daris Chen

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Education

University of California San Diego
Bachelor of Science (B.S.) in Computer Engineering — *San Diego, CA*

Work Experience

Lead Full-Stack Engineer | PatentIQ — Jan 2026 – Apr 2026
Product Manager Accelerator — *Remote*

- Orchestrated Agile development workflows via ClickUp for a 14 member cross functional team while serving as lead architect and primary contributor to deliver the foundational PatentIQ AI technical MVP.
- Constructed an AI-powered patent search and drafting assistant hosted on Vercel, EC2, and Supabase using Next.js, Auth0, and FastAPI to simplify intellectual property research for early-stage founders.
- Programmed a hybrid semantic search engine combining OpenAI text-embeddings and PostgreSQL pgvector to perform high-dimensional similarity ranking and recommendation system over global USPTO patent datasets.
- Built recursive SQL query builder with Zod-validated JSON filters and implemented an LRU cache to DB reduce latency.

Lead Full-Stack Engineer — Aug 2025 – Mar 2026
Groundwork Books — *San Diego, CA*

- Led architectural migration of production e-commerce to Next.js and Square for live inventory synchronization.
- Engineered a semantic search layer using Pinecone with integrated text embeddings, enabling topic and concept-based discovery across 4,000+ SKUs without requiring exact title matches.
- Architected a two-tier caching system: a look-aside Redis reducing Square API latency by 90% (sub-200ms p99), and a client-side request coalescing hook batching 50+ concurrent inventory lookups into an API call with 2-minute TTL cache.

FishSense Artificial Intelligence Researcher — Feb 2025 – Mar 2026
UCSD Engineers for Exploration (E4E) — *San Diego, CA*

- Engineered a custom R-CNN training pipeline (ResNet-50 backbone) for underwater laser-profiling, standardizing the Detectron2 codebase, to reducing inference setup time by 40% for new researchers and increasing reproducibility.
- Developed a dual-laser photogrammetry algorithm to extract biological metrics (fish length/mass) from 2D video feeds, achieving sub-5% error rates in pixel-to-centimeter conversion.

Software Engineering Intern — Jan 2025 – Sep 2025
Personify AI (Pre-Seed EdTech) | personifyapp.ai — *Remote*

- Refactored feature leaderboard user interface using Next.js, Tailwind CSS, and Firebase, increasing user traffic by 15%; designed duplicate-detection logic in Firebase to clean the data pipeline.
- Built admin controls for approvals, rejections, and closures, and added vote-based ranking, category sorting, and search, cutting resolution time by 50% and reducing user interaction time by 20%.
- Engineered a data ingestion pipeline for unstructured text, implementing deduplication logic and NLP-based feature extraction to structure dataset inputs for the recommendation engine.

Projects

Phased Array Microwave Sensor | *ESP32-S3-DevkitC-1, Raspberry Pi 4, Analog Design* — Feb 2026 – Apr 2026

- Constructed an analog front-end for a 4x1 array of HB100 microwave transceivers implementing a dual-stage op-amp cascade (MCP6002) to achieve a 1092x gain for microvolt intermediate frequency signals and 1.65V virtual ground biasing to process bipolar IF signals on a unipolar ADC.
- Programmed a processing pipeline combining 4th-Order Butterworth filters and FFTs with MUSIC algorithms and an Extended Kalman Filter for precise direction estimation and 2D spatial tracking.
- Coded an amplitude monopulse zone classifier to calculate spatial bias and target velocity substituting phase-coherent direction finding to overcome the inherent local oscillator drift of the hardware.

TriCore9 | *SystemVerilog, Assembly, Python, ModelSim, Quartus Prime* — Sep 2025 – Dec 2025

- Architected a custom 8-bit RISC soft-core processor with a 4-stage pipeline (Fetch, Decode, Execute, Writeback), optimizing data paths to minimize hazard stalls.
- Engineered a custom Assembler in Python to translate mnemonics to machine code, enabling rapid validation of the ISA through complex Hamming Code algorithms.

16 bit Carry Look Ahead Adder | *Verilog, Digital Logic Design (Digital)* — Sep 2024 – Dec 2024

- Constructed a hierarchical 16-bit adder using cascaded 4-bit PG blocks to reduce critical path delay to $O(\log N)$, implementing 2421 BCD variants validated through corner-case simulations.

Technical Skills

Programming Languages: Python, Java, C/C++, JavaScript/TypeScript, HTML/CSS, ARM Assembly, SQL, Verilog/SystemVerilog, Go, Cython
Libraries & Frameworks: PyTorch, TensorFlow, CUDA, ROCm, cuDNN, Transformers, NumPy, pandas, Matplotlib, Scikit-learn, Detectron2, tqdm, React, Next.js, Node.js, Express.js, Three.js, Svelte, Tailwind CSS, Zod, DepthAnythingV2, DFormerV2, SAM3D Object, OpenAI, Claude, Signal Processing, FFT
Tools & Technologies: Git, GitHub Actions, Docker, Kubernetes, Linux, Clickup, Jira, Jest, Puppeteer, GCP, gRPC, EC2, GCC, OpenCL, Figma, VS Code, CodeMirror, Wandb, Jupyter, Hugging Face, Vercel, ESLint, Turbopack, OSM Nominatim, Open-Meteo, OpenStreetMap, Intel FPGA ModelSim, WebSpatial (Vision Pro), Auth0, SSH, HB100, RF Engineering, ESP32
Databases: PostgreSQL, MongoDB, Upstash Redis, Firebase, Pinecone, pgvector, Supabase